

ECON 1001 – Midterm 1 – 1 October 2025

Choose the best response based on the models we have studied in class. Each problem or subproblem is worth 4 points.

- 1) Consider the market for Crocs. In 2015, $P = 30$ and $Q = 1,000$. In 2025, $P = 70$ and $Q = 50,000$. Which of the following is true?
 - a) Inflation caused the price increase between 2015 and 2025.
 - b) These are two points on an upward-sloping demand curve.
 - c) These are two points on different downward-sloping demand curves.
 - d) The supply curve for Crocs shifted out, causing the price and quantity to increase.
- 2) “The Trump administration’s tariffs are bad.” This statement is:
 - a) Positive and true
 - b) Positive, though we can’t say if true or false.
 - c) Normative
 - d) Negative
- 3) Amanda’s favorite band was coming to town. Amanda knew the line to buy a \$20 ticket would take two hours and the value of her time waiting in line is \$25/h. Unfortunately, when she got to the front of the line, the \$20 tickets were sold out (none were left). However, there was a \$50 ticket for a seat in a better location (Amanda preferred that location). Amanda, who also acts rationally, would
 - a) Not buy the \$50 ticket.
 - b) Buy the \$50 ticket.
 - c) Call the \$20 ticket a sunk cost.
 - d) [We can’t say what she’d do because we don’t know enough about Amanda’s WTP for the better seat.]
- 4) Suppose at the beginning of the fall semester, you paid \$5,000 for an unlimited meal plan at the cafeteria, which permits you to enter the cafeteria at any time to eat a meal. The marginal financial cost to you of eating a meal (that is, what you’d have to pay in dollars to eat there) at the cafeteria is
 - a) Zero.
 - b) Depends on how many times you eat at the cafeteria this semester.
 - c) Depends on how much you like the food.
 - d) Depends on how much a meal costs at an equally convenient location.

- 5) The production possibility frontier in Figure 1 (last page) shows how many additional points Lucien can get on his economics and psychology exams based on how he spends his time studying. The PPF
- a) Indicates the optimal number of hours for Lucien to study for psychology.
 - b) Shows the tradeoff Lucien faces in studying psychology versus economics.
 - c) Doesn't make any economic sense.
 - d) Shows Lucien needs to study for both classes.
- 6) Nate bought a trumpet from Caleb for \$50, even though Nate's WTP for the trumpet was \$75. Thus
- a) The total economic surplus from this transaction was exactly \$25.
 - b) Caleb got a bad deal.
 - c) Caleb enjoyed less economic surplus from this sale than Nate.
 - d) The total economic surplus from this transaction was at least \$25.
- 7) The cost-benefit principle is that "an individual does something only if the benefit of doing that thing is greater than (or equal to) the cost of doing it." [As usual, assume the individual has accurate information about the costs and benefits of their chosen action.] The cost-benefit principle ...
- a) Does not apply to decisions like going to visit your sick grandmother.
 - b) Depends on whether demand is elastic.
 - c) Only applies to business decisions.
 - d) Leads a person to maximize their economic surplus.
- 8) Ewa buys widgets in the perfectly competitive market for widgets. When Ewa's income goes up, *ceteris paribus*,
- a) She buys fewer widgets.
 - b) She buys more widgets, but we can't say how many more without knowing how the price changes.
 - c) What happens depends on whether there are any close substitutes to widgets.
 - d) She adjusts how many widgets she buys, depending on her income elasticity of demand.
- 9) On \$1 Milkshake Day at that one milkshake shop near campus, the quantity of milkshakes bought and sold goes way up (compared to a typical day when the price is \$7). That is explained
- a) As a shift in the market demand curve for milkshakes, because so many more people than usual go to the shop.
 - b) By a fall in the cost to the shop of producing milkshakes.
 - c) As a movement along the market demand curve.
 - d) By the milkshakes sold being smaller than usual.

- 10) There is a new good called *Two-Year Tattoos*. Two-Year Tattoos are just like traditional tattoos, except they immediately disappear after two years. Some suppliers of traditional (permanent) tattoos start providing Two-Year Tattoos instead of traditional tattoos. As a result of the invention of Two-Year Tattoos, what happens in the market for traditional permanent tattoos?
- a) The market for two-year tattoos has no effect on the market for traditional tattoos.
 - b) The equilibrium quantity and price of traditional tattoos both fall.
 - c) The equilibrium quantity falls, but the effect on the equilibrium price depends on the relative size of the change in demand and supply.
 - d) The price falls but the quantity stays the same.
- 11) Rebekah sells lemonade in a *perfectly competitive* market where $P = 3$.
- a) If Rebekah works harder, she could lower her price a bit (say to 2.5) and make more money.
 - b) Rebekah could increase the price she charges, so long as demand is elastic at $P = 3$.
 - c) She sells the quantity on her supply curve that goes with $P = 3$.
 - d) Even though there is a large quantity demanded at $P = 3$, Rebekah doesn't sell more than her fair share of lemonade because she doesn't want to face increased competition from her competitors.
- 12) Rebekah sells 100 cups of lemonade in a perfectly competitive market where $P = 3$. Suppose Rebekah's fixed costs increase from 5 to 10. Then, *ceteris paribus*,
- a) Rebekah will sell fewer than 100 cups to make up for the higher fixed costs.
 - b) Rebekah will sell more than 100 cups to make up for the higher fixed costs.
 - c) Rebekah will continue to sell exactly 100 cups.
 - d) Rebekah is completely unaffected by the increase in fixed costs: her profits will be the same.
- 13) Suppose you manage a Cookie Shop (in the perfectly competitive market for cookies), and you employ only high-school students. When the equilibrium wage for high-school workers increases,
- a) You move down your supply curve, supplying fewer cookies.
 - b) You move up your supply curve, because the price has increased.
 - c) Your supply curve shifts out and you sell more cookies.
 - d) Your supply curve shifts in and you sell fewer cookies.
- 14) You run a Sandwich Shop (in a market that is not perfectly competitive). Your economist tells you that at the current price your customer's price elasticity of demand is elastic. If you lower the price a little,
- a) Total revenue will fall but total profits will increase.
 - b) Total revenue will fall and total profits will fall.
 - c) Total revenue will increase and total profits will increase.
 - d) It's unclear whether total profits would increase or fall or stay the same.

- 15) Suppose the mayor in a certain city can choose Policy A, which would fix the bus station in the city center. Policy B would fix bus stops on the edges of town. Policy A would increase economic surplus by \$100: \$80 for Rich People, \$20 for Poor People. Policy B would create economic surplus of \$60: \$30 for Rich People and \$30 for Poor People. Suppose the mayor's sole concern is to help Poor People.
- He should choose Policy A because it is more efficient.
 - He should choose Policy B because it is more equitable.
 - If the mayor has the power to redistribute economic surplus without destroying economic surplus, then he should choose Policy A. If cannot redistribute, then he should choose B.
 - It depends on whether the equity-efficiency tradeoff applies here.
- 16) Based on *the bottom two rows* in Rebekah's supply curve in Table 1, what is the arc elasticity? [You do NOT need to check my arithmetic.]
- 1
 - $\frac{\frac{1}{2.5}}{\frac{1}{10.5}} = 4.2$
 - $\frac{\frac{1}{2}}{\frac{1}{10}} = 5$
 - $\frac{\frac{1}{3}}{\frac{1}{11}} \approx 3.67$
- 17) Over a period of time in Hannahland, the price of electric hybrid vehicles decreased while the number of electric hybrid vehicles sold increased. What happened?
- There was an increase in demand (i.e., the demand curve shifted out).
 - There was definitely an increase in supply AND an increase in demand (i.e. both curves shifted out).
 - There was definitely an increase in supply, though it's unclear whether the demand curve shifted in or out.
 - The price of gas-powered vehicles was too high.
- 18) Suppose the market for widgets is perfectly competitive (and the markets meet all the other right conditions). Total economic surplus is maximized if
- Each widget supplier has the same *total costs*.
 - Each demander who purchases a widget has the same *total benefits*.
 - Suppliers and demanders are not overly greedy.
 - [None of the above.]
- 19) Suppose the mayor wants to impose a \$5-per-unit tax that affects consumer surplus and producer surplus equally. Which of the following accomplish that?
- A \$2.50-per unit tax on demanders and a \$2.50-per unit tax on suppliers.
 - A "sin tax" on a good like cigarettes, alcohol, or unhealthy foods.
 - A \$5-per-unit tax on a good whose elasticity of supply equals the (absolute value of) the elasticity of demand.
 - [None of the above.]

20) In Figure 2 (last page), what is the economic surplus at the competitive equilibrium?

- a) $A + B + C + F + G + H + K + L + M$
- b) $A + B + C + F + G + K$
- c) $A + B + F + K$
- d) $A + B + C$

21) In Figure 2 (last page), what is the producer surplus when there is a quota of 15?

- a) $A + B + C$
- b) $F + G + K$
- c) $B + F + K$
- d) $B + C + F + G + K$

22) In Figure 2 (last page), what is the deadweight loss if the government imposes a price ceiling of 13?

- a) $D + E$
- b) $C + G$
- c) $A + C + G$
- d) Can't say: It depends on whether the price ceiling is binding.

23) Bubble in **A**. (This indicates the version of your exam.)

Free Response – Respond on the back of your bubble sheet.

24) Supply and demand for widgets are $Q_D = 5.5 - .5P$ and $Q_S = P - 2$.

- a) Diagram the situation. Be sure to label the supply curve, the demand curve, and the intercepts on the vertical axis (i.e., where the curves hit the price axis).
- b) What is the price at the equilibrium?
- c) Suppose the government imposes a price floor of $P = 9$. How many widgets will actually be sold?
- d) Suppose that instead of a price floor, the government imposes a tax of \$3 per unit. What will the quantity Q_t be? (i.e., the quantity with the tax.)

25) The following table shows Ruari's **total benefits** associated with consuming various quantities of cookies. Fill in Ruari's demand table on your response sheet. [Assume that Ruari can only purchase whole numbers of cookies. That is, there are no fractional cookies. Also remember our class convention: if $MB = MC$, she buys it.]

Q	Total Benefit
1	10
2	18
3	24
4	28

DO NOT TURN THIS OVER UNTIL INSTRUCTED TO. Good luck!!

Table 1: Rebekah's Supply

P	Q
9	1
10	2
11	3

Figure 1: Lucien's PPF

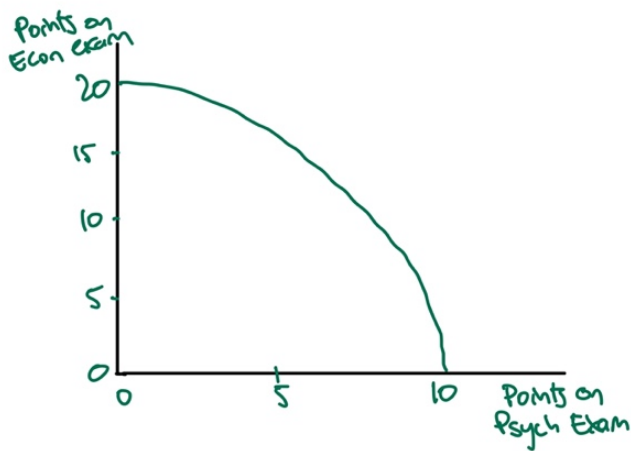


Figure 2: Surplus Shapes

